

JANUARY 2018

Disruptors, Innovators, and Thieves

Assessing Innovation in China's Digital Economy

AUTHOR
Samm Sacks

CSIS | CENTER FOR STRATEGIC &
INTERNATIONAL STUDIES

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China Innovation
Policy Series

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The China Innovation Policy Series

The purpose of the China Innovation Policy Series (CIPS) is to analyze trends in technology innovation in China and consider the implications for government policies and business strategies. The series examines general trends as well as a range of strategic sectors that have different economic dynamics, including: the Internet, new-generation vehicles (electric and autonomous vehicles), semiconductors, artificial intelligence, commercial aircraft, and pharmaceuticals. Over the life of the two-year project, reports are being issued on each sector, and CSIS is hosting events to discuss these reports and the broader issues that will shape China's innovation path and how others in government and industry should respond.

CIPS is a joint project of the Freeman Chair in China Studies and the Technology Policy Program at CSIS. The principal investigators are: Scott Kennedy, deputy director of CSIS's Freeman Chair in China Studies and director of the Project on Chinese Business and Political Economy; Samm Sacks, senior fellow in CSIS's technology policy program; and James Lewis, senior vice president at CSIS.

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Introduction

The Chinese leadership has made clear that the digital economy and information and communication technologies (ICT) now underpin the country's economic development. President Xi Jinping's slogan that "without informatization there is no modernization" captures the government's goal to use internet technology to unlock greater productivity in the real economy. There is high-level attention on merging online with offline through development of mobile, artificial intelligence (AI), big data, cloud, and Internet of Things (IoT). This focus is not new but goes back to at least 2015 when the government released an initiative called Internet Plus, calling for targeted support to develop these sectors. Yet, the term "digital economy" is appearing more frequently just since the 19th Party Congress in October 2017, when President Xi called out the term for the first time ever in an opening Party Congress Speech.¹

This is not just party rhetoric coming from the top. In the private sector, global investors are describing China's private sector as "futuristic" and teeming with "dynamism, creativity, and entrepreneurship."² China's behemoth internet companies (Baidu, Alibaba, Tencent, JD.com—collectively referred to as "BATJ") are growing more influential in and outside of China. Meanwhile, China's startup scene is thriving as entrepreneurs create new technologies and business applications that make the digital economy an integral part of daily life.

Consensus is forming around two main points. The first is that China's internet companies are no longer mere "copycats" or clones of Western companies—or as one observer in Beijing put it: "Anyone who says that China is not innovative or entrepreneurial is kidding themselves."³ The second is that BATJ are making a strong push to expand to global markets. Both are well-documented, along a well-trodden path: Are Chinese companies true innovators or just localizing technology and business applications from successful Western companies?

1. We thank Paul Triolo and Jennifer Meng for observing the first use of "digital economy" in an opening Party Congress speech.

2. Interview with Shanghai-based venture-capital firm, October 2017, and Hans Tung, December 2017.

3. Lilian Rogers, "China's Tech Titans, from Another Planet," U.S. Information Technology Office, September 22, 2017, <https://www.linkedin.com/pulse/chinas-tech-titans-from-another-planet-lilian-rogers>.

Will Chinese internet companies that dominate China be able to capture global market share or are they only succeeding because of protectionism in a closed market? Indeed, strong arguments for each side can be found in writings throughout Chinese and Western media, think tanks, and industry outlets.

Our report does not seek to rehash these arguments; instead, we reframe the discussion to focus on three findings:

First, a case can be made to show that China's internet companies are at once disruptive innovators, micro-innovators, and copycats. All types exist, making the ongoing question—is China innovating or not?—less relevant.

Second, we are seeing a phenomenon of “cross border convergence,”⁴ in which trends in China's digital economy are increasingly shaping U.S. and global companies, and vice versa. Will Wechat ever capture real market share beyond China? Probably not (although some will argue this point). But the more interesting issue is the way in which Wechat's super app or “dynamic integration” (有机结合) model⁵ is influencing new features and applications being developed outside of China. This cross-border “convergence” runs both directions, as BATJ pour money into research and development (R&D) of cutting-edge fields, learning from technological advances and experts from Western economies.

Third, there are new policy challenges that could get in the way of cross-border convergence around innovation. China's emerging cybersecurity regime (with China's cybersecurity law as the center) runs the risk of contributing to a trend that could lead to a semi-balkanized internet. The risk is that some companies operating in China will face restrictions next year in their ability to move data critical to business operations and optimizing data analytics freely across borders. At the same time, China's ICT sector is becoming more closed to foreign competition for network products and services. Localization of China's ICT sector means that China's internet companies 1) lose opportunities to become more internationally competitive; and 2) may face new restrictions in global markets as policymakers in the United States and possibly Europe look to exert pressure on Beijing for reciprocity.

WHAT DOES “INTERNET” AS A SECTOR ACTUALLY MEAN?

The scope of this study focuses on internet companies and the internet sector in China. But the terms needs clarification since in reality, Chinese “internet” companies are expanding into many more areas in which internet is more accurately understood as an underlying technology applying to subsets of the broader ICT sector.

4. Han Tung, phone interview with the author, December 2017; and Austin Car, “These Six Chinese Tech Giants Are Ramping up the Pace of Innovation,” Fast Company, February 13, 2017, <https://www.fastcompany.com/3067467/these-six-chinese-tech-giants-are-ramping-up-the-pace-of-innovation>

5. 刘一鸣 (Liu Yiming), “中国互联网公司进入大航海时代:海外扩张成下一轮机遇” (“China's Internet Companies Enter the Era of Big Sails: The Next Round of Opportunities for Overseas Expansion”) 财经, September 5, 2016, <http://tech.sina.com.cn/i/2016-09-05/doc-ixvqcts9494640.shtml>.

This report focuses on “Internet Plus”—a term linked to an initiative released in 2015. The plan refers to mobile internet, cloud, big data, and internet of things (IoT) integrated with traditional industries and manufacturing. Or, as one senior official at a leading Chinese internet firm described it:

The Internet part of “Internet Plus” refers to artificial intelligence (AI), big data, and cloud; IoT is the “plus.” This is like the brain and the nerves, in which IoT refers to everything the internet touches such as manufacturing, logistics (including transportation), and payments.⁶

In particular, the Chinese government and industry are focused on using internet technologies to unlock the productivity of the real or conventional economy, a concept known as online to offline (O2O), such as ecommerce, ride hailing, and online delivery services.⁷ Leading China high-tech venture investor Kai-fu Lee also calls this OMO (online merges with offline), and claims that China is leading in this merging of the internet economy with the traditional economy.⁸ For example, the Wuzhen Outlook published right after the World Internet Conference in Wuzhen in December defines digital economy as “represented by e-commerce, sharing economy, digital inclusive finance, and industrial internet,” and calls for strengthening “integration between the digital and real economy.”⁹ This is echoed a similar formulation in Xi Jinping’s work report to the Party Congress in October, and will be a continuing refrain throughout 2018.

Our report, therefore, uses the term “internet companies” for simplification, but with an understanding that the concept and wording are more complex, and in a sense, the term is outdated—terms like “platforms” to describe the BATJ provide more clarity. Only the United States and China have major internet companies that are more like “platforms”—there is no real equivalent in Europe or anywhere else in the world. A decade ago the distinction between software, hardware, and internet companies may have made more sense.¹⁰ An executive at one of the BATJ companies, for example, commented that AI is integrated into business operations across the company making services from logistics and payments more efficient.¹¹ Cloud is also a tool that enables companies across technology sectors to innovate by storing, transferring, and computing operations performed on data. Large companies such as Alibaba are using cloud for their own services like Taobao and Alipay, at the same time they are commercializing cloud for third-party customers using software based on a cloud platform.¹²

The bulk of this report focuses on the over-the-top (OTT) layer of the internet (which refers to content, services, and applications delivered over the internet), rather than the internet service

6. Interview with Chinese internet company executive, December 8, 2017.

7. Julia Wang, “China Outperforms on Innovation,” HSBC Global Insights Briefing, October 13, 2017, http://www.gbm.hsbc.com/insights/global-research/china_outperforms_on_innovation.

8. Kai-Fu Lee and Jonathan Woetzel, “China, the Digital Giant,” Project Syndicate, December 4, 2017, <https://www.project-syndicate.org/commentary/china-digital-age-innovation-by-kai-fu-lee-and-jonathan-woetzel-2017-12>.

9. “Wuzhen Outlook,” World Internet Conference Wuzhen Summit, December 6, 2017, http://www.wuzhenwic.org/2017-12/06/c_117171.htm.

10. Interview with Paul Triolo, December 27, 2017.

11. Interview with Chinese internet company, December 2017.

12. Email interview with U.S. ICT company representative in Beijing, December 2017.

providers (ISPs)/operators responsible for the network infrastructure. It also looks at some of China's equipment makers such as Huawei. Quantum computing and encryption are outside the scope of the report. In addition, this report does not delve deeply into semiconductors and AI; both will be the subject of their own CSIS reports coming later in 2018. AI is an increasingly important part of internet company and platform development, and is being integrated into a range of business operations of these firms.¹³ As a factor in assessing China's overall level of high-tech innovation, the complexities of AI require its own framework for analysis.

STRUCTURE

The report consists of three sections:

- The first section of our report lays out Beijing's vision for the internet and digital economy—with the understanding that just because this is what Beijing wants does not mean this will be realized.¹⁴
- Second, we look at what Chinese internet firms are actually doing. We lay out the three narratives about innovation in China's internet economy: Disruptor, Micro-innovator, and Copy to China. We give brief case studies to show that all exist. For each, we address the issue of Chinese internet companies going global. The argument that Chinese internet companies cannot succeed both in China's market and overseas is already well-documented,¹⁵ but we offer a few up-to-date illustrations of this, as well as why we think cross-border convergence is a more important lens through which to view the issue.
- The third section assesses policy challenges, focusing on the impact of restrictions on cross-border data flows as well as the implications of ICT localization in China.

13. Kai-Fu Lee and Paul Triolo, "China's Artificial Intelligence Revolution: Understanding Beijing's Structural Advantages," Eurasia Group, December 2017, https://www.eurasiagroup.net/files/upload/China_Embraces_AI.pdf.

14. National policy is the first of three "operational levels" that Jerry Haar and Ricardo Ernst identify as playing a role promoting innovation in *Innovation in Emerging Markets* (New York: Palgrave MacMillan, 2016). The others are facilitating institutions (i.e., universities, incubators, labs, and industry associations), and firm-level behavior (products, services, processes, and business models).

15. Liu Yiming, <http://tech.sina.com.cn/i/2016-09-05/doc-ifxvqcts9494640.shtml>.

National Policy: Beijing's Vision for Internet Technology and the Digital Economy

Some scholars argue that the Chinese government contributes to technological leadership of Chinese companies.¹⁶ Regardless of controversy¹⁷ surrounding state-directed technology development, it is important to understand the Chinese government's objectives.

Dozens of state plans and speeches boil down to three main objectives: 1) strengthen the digital economy in order to enhance economic development; 2) encourage the growth of globally influential internet companies that will drive advancements in next-generation ICT fields like AI, big data, cloud, and IoT; and 3) build out a robust legal framework to strengthen party control over emerging technologies. Underlying these objectives, Beijing sees the development of ICT as crucial for moving up the value chain to increase productivity in the Chinese economy. These goals underpin a concept that the leadership refers to as "building a national power in cyberspace," also translated as "building China into a cyber superpower."¹⁸

WHAT IS THE GOVERNMENT LOOKING TO ACHIEVE?

The clearest explanation of President Xi's goals appears in an essay published in September by the Cyberspace Administration of China (CAC) Theoretical Research Institute published in *Qiushi*, a top party journal. This essay should be understood as an authoritative statement coming from the senior leadership explaining policy plans (rather than an outside academic commentary):

16. Can Huang and Naubahar Sharif, "Global Technology Leadership: The Case of China," *Science and Public Policy* 43, Oxford University Press, 2015.

17. CSIS, "China's Power: Up for debate," December 7, 2017: *In the coming decade, China will emerge as a global leader in technology innovation*, with Huang Can and Adam Segal, available at <https://www.youtube.com/watch?v=Fxpe-J1nYy9A>.

18. Elsa Kania, Samm Sacks, Paul Triolo, and Graham Webster, "China's Strategic Thinking on Building Power in Cyberspace: A Top Party Journal's Timely Explanation Translated," *New America*, September 25, 2017, <https://www.newamerica.org/cybersecurity-initiative/blog/chinas-strategic-thinking-building-power-cyberspace/>.

*"Make the Internet and information industry both big and strong. Persist in coordinating the development of support with that of rules (坚持鼓励支持和规范发展并行), in the simultaneous development of policy guidance with laws and supervision, creating a favorable market environment for the development and building of the Internet. The global influence of Internet companies like Alibaba, Tencent, Baidu, Huawei, etc., is on the rise. One after the other, many innovative and vital Internet companies have emerged, playing an important role in promoting stable growth and employment, and benefitting the people. In 2016 on a global list of top 20 companies by market value, Chinese companies occupied seven slots."*¹⁹

Indeed Chinese internet companies are expanding their global footprints. The year Tencent bought stakes in 15 companies at \$4.3 billion, Alibaba had eight cross-border deals worth \$2.4 billion.²⁰ These global investments align with the government's strategic technology development plans while also providing opportunities to learn from major economies and gain greater bargaining clout.

Just weeks after the *Qiushi* essay came out, President Xi gave his opening speech at the 19th Party Congress. This speech is important because it sets the policy priorities for the next five years of Xi's term with the highest authority.²¹ It is not new or notable that the speech mentioned innovation or informatization, since these have been buzzwords for years. However, it is the first time that an opening speech identifies specific terms such as AI and digital China as priorities in the country's development plans. For example, the speech calls for accelerating the "deep integration of the Internet, big data, and AI with the real economy." It calls for building a "science and technology superpower, quality superpower, aerospace superpower, cyber superpower" and "intensifying cooperation in frontier areas such as digital economy, AI, nanotechnology, and quantum computing, and advancing the development of big data, cloud computing, and smart cities so as to turn them into a digital silk road of the 21st century."²² The last point, while party rhetoric, does indicate the direction of government planning.

Xi's speech also indicated how the leadership views the digital economy within the broader context of economic policy. The speech does not repeat the target of doubling per capita gross domestic product (GDP) by 2020 from the last Congress. Deemphasizing GDP and focusing instead on what the Chinese government calls "supply side structural reforms" (i.e., boosting high-tech sectors rather than traditional heavy industry) implies an ideological shift in the Chinese Communist Party (CCP), with the leadership prioritizing higher-quality drivers of economic development.²³

19. Ibid.

20. Louise Lucas, "Tencent and Alibaba Go Abroad to Push for Growth and Know-How," *Financial Times*, December 15, 2017, <https://www.ft.com/content/a968cf62-de3f-11e7-a8a4-0a1e63a52f9c>.

21. Alice Miller, "How to Read Xi Jinping's 19th Party Congress Political Report," Hoover Institution, May 25, 2017, <https://www.hoover.org/research/how-read-xi-jinpings-19th-party-congress-political-report>.

22. 习近平作十九大报告全文实录 (Full text of Xi Jinping's report at 19th CPC National Congress), October 18, 2017, <http://news.163.com/17/1018/15/D11S5V3Q0001899O.html>.

23. Sam Sacks, "Cyber Policy and the 19th Party Congress," CSIS Critical Questions, October 26, 2017, <https://www.csis.org/analysis/cyber-policy-and-19th-party-congress>.

HOW IS BEIJING CARRYING OUT THIS VISION?

Beijing issued a number of national strategies, policies, and plans for advanced ICT and the digital economy from the last several years. These should not be dismissed as mere aspirations or empty party slogans. Taken together, these directives represent a detailed blueprint for state support across specific sectors. Foreign and domestic companies, investors, research institutes, and universities have financial and policy incentives to align with the blueprints articulated in these directives:

Guiding Opinions on Actively Promoting Internet Plus (July 2015)²⁴

- Calls for integrating AI, big data, cloud into the “real economy” via the IoT
- Focus on online to offline (O2O) (i.e., ecommerce, ride hailing, food ordering) to unlock “hidden productivity of the conventional economy”

13th Five-Year Plan for Informatization (2016–2020)²⁵

- Aims for China to “No longer be restrained by others [i.e., foreign industry/governments] for core technologies in strategically competitive fields”²⁶
- Identifies major projects slated for increased state support in “core electronic equipment, high-end general-purpose chips, basic software, large-scale IC, next-generation wireless broadband mobile communication, quantum communication and quantum computing.”
- Calls for building a “National Data Resource” system to enhance coordination of data

13th Five-Year Plan for Science & Technology Innovation (August 2016)²⁷

- Sets R&D spending targets and calls for breakthroughs in next-generation ICT like cloud and big data
- Support Chinese firms to set up overseas R&D centers while encouraging foreign firms and research institutions to set up high-end R&D centers for technology transfer

National Informatization Development Plan (July 2016)²⁸

- 10-year roadmap for boosting China’s ICT industry globally

24. 国务院关于积极推进“互联网+”行动的指导意见 (Guiding Opinions on Actively Promoting Internet Plus), July 4, 2015, http://www.gov.cn/zhengce/content/2015-07/04/content_10002.htm.

25. 国家信息化“十三五”重点规划 (“13th Five-Year Plan for Informatization (2016–2020),” December 27, 2016, http://www.gov.cn/zhengce/content/2016-12/27/content_5153411.htm.

26. The comment in brackets is the authors’ and not in the original document.

27. 十三五”国家科技创新规划 (“13th Five-Year Plan for Science & Technology Innovation”), August 8, 2016, http://www.gov.cn/zhengce/content/2016-08/08/content_5098072.htm.

28. 国家信息化发展战略纲要 (“National Informatization Development Plan”), July 28, 2016, <https://chinacopyrightandmedia.wordpress.com/2016/07/27/outline-of-the-national-informatization-development-strategy/> and http://news.xinhuanet.com/info/2016-07/28/c_135546104.htm.

13th Five Year National Strategic Emerging Industries Development Plan²⁹

- Developments in next generation ICT (AI, cloud, big data, IoT) as source of growth and “symbol of national strength”

Belt and Road Initiative³⁰ and Information Silk Road

- Calls for promotion of ecommerce, smart cities, and digital trade with Asia, Europe, and Africa, with expansion of connectivity and communication networks

Broadband China Strategy (August 2013)³¹

- Aims to improve internet speed, coverage, and services across the country through targeted infrastructure investments

WHAT THE NUMBERS TELL US ABOUT WHY BEIJING IS SO FOCUSED ON THE DIGITAL ECONOMY

The high-level attention by China’s leaders on the internet and digital economy reflects the sheer scale of China’s digital landscape. To give an idea of why Beijing sees such massive potential for growth, China has 751 million internet users— one-fifth of the global total. But more importantly, the fact that China’s internet penetration rate stands at 54.3 percent, compared with 84 percent in the United States, suggests there is more upside yet to be captured.³² Smartphones are the centerpiece of O2O/OMO applications, with devices as the lynchpin for digital payments, ride sharing, and more. This means that growth of mobile internet users in particular is an important piece of China’s digital economy story, especially for the growth of internet finance. China’s mobile users (over 95 percent of internet users) have been growing at an annual growth rate over 10 percent for the past three years. Mobile online payment in particular increased over 30 percent from 2016 to 2017; half of China’s internet users paid for products and services on their phones.³³

It is also no surprise that the leadership is increasingly focused on the global influence of China’s large internet companies given the political, economic, and security implications. Appendix A shows how China’s top internet companies stack up in comparison with those in the United States in terms of revenue, market share, and R&D spending (in absolute terms and as a percentage of revenue).

29. “十三五”国家战略性新兴产业发展规划 (“13th Five Year National Strategic Emerging Industries Development Plan”), http://www.ndrc.gov.cn/fzgggz/fzgh/ghwb/gjjgh/201705/t20170508_846776.html.

30. “Vision and Actions on Jointly Building Silk Road Economic Belt and 21st-Century Maritime Silk Road,” Belt and Road Initiative White Paper, March 2015, http://en.ndrc.gov.cn/newsrelease/201503/t20150330_669367.html; and Digital Belt and Road Initiative Held in Beijing, http://english.radi.cas.cn/News/NU/201612/t20161221_172699.html.

31. 国务院关于印发“宽带中国”战略及实施方案的通知 (State Council Notice Concerning “Broadband China” Strategy and Implementation Plan), http://www.gov.cn/zwggk/2013-08/17/content_2468348.htm.

32. 40th China Statistical Report on Internet Development released by the China Internet Network Information Center (CNNIC), January 2017, <https://cnnic.com.cn/IDR/ReportDownloads/201706/P020170608523740585924.pdf>.

33. CCNIC and Zennon Kapron, “Finance Disrupted: How digital finance is changing the way 1.3 billion people bank and the implications for US-China relations,” U.S.-China Economic and Security Review Commission, June 21, 2017.

Disrupters, Micro-innovators, and Thieves

There are three main storylines about innovation in China's internet sector. There are cases that illustrate the way in which China's internet companies are disruptive innovators, micro-innovators, and clones. All types exist, making the ongoing question—is China innovating or not?—less relevant. Instead, we look at what China's leading internet companies are doing to shake up the industry through integrating digital technologies more seamlessly into business operations and consumer daily life—merging online and offline. This section looks at how these three models are playing out inside China, then assesses their prospects globally.

Narrative 1: Disruptive Innovation

Disruptive innovation creates new markets and displaces existing technologies and business models.³⁴ Sometimes this kind of innovation involves no disruption in core technology, but is more about transformation in terms of application or business strategies to find new solutions to real-world problems. We highlight three examples where we see this kind of disruptive innovation taking place: internet finance and the Chinese lip-syncing video app Musical.ly are two examples on the OTT side. Huawei's is an example of an equipment maker that is innovating in management strategy and industry approach in ways that will reshape the next-generation internet.

1. Internet Finance

China has one of the most advanced and largest internet finance market ("fintech") in the world.³⁵ China's fintech platforms are revolutionizing the way people manage their finances and make payments. With a mobile payments markets over \$5.5 trillion (roughly 50 times the size of that in the United States), a cashless life in China seems more possible than in any other country. Mobile payments are ubiquitous in China from taxis, restaurants, convenience

34. Haar and Ernst, *Innovation in Emerging Markets*, 21.

35. We thank Martin Chorzempa, of the Peterson Institute of Economics, for sharing his insights on China fintech with me through several conversations from November to December 2017.

stores, and retail to sidewalk hand crafts.³⁶ Millions of stores accept digital payments by scanning QR codes. Fintech in China does not just consist of digital payments, but entire “one-stop financial service ecosystems” bringing together digital payments, financial credit, lending, investments (i.e., wealth management and stocks) with ecommerce, transportation, entertainment, and travel.

Platforms like Ant Financial, the financial affiliate of Alibaba, and Tencent’s WeChat Pay are disrupting the financial sector in China. They have no real Western equivalents.³⁷ Comparisons to companies like Paypal do not hold up. Ant Financial, for example, is valued at over \$70 billion with its own online bank, lending arms, and one of the largest money market funds in the world, Yu’e Bao. In fact, the rapid growth of these platforms has led Chinese government regulators to try to reign in online finance through several waves of restrictions over the past several years. China’s slower-moving incumbent banks have the most to lose from the rise of online finance, which has been a dominant force behind much of the regulatory measures.

Technology is a major enabler of China’s fintech sector. The sheer scale of transactions that Alibaba and Tencent process dwarfs that in the United States. For example, on Singles Day in China this year (where Alibaba had over \$25 billion in sales this year, compared to \$12.8 billion in the United States between Thanksgiving and Cyber Monday in 2016), Alibaba processed 256,000 transactions per second and handled 1.48 billion transactions in 24 hours.³⁸ According to Tencent’s annual report, its payment infrastructure “enabled us to process peak volume of 760,000 red packets (electronic cash gifts traditionally given in red envelopes) during the Lunar New Year.”³⁹ By contrast, Visa states on its website that it can handle 24,000 transactions per second.⁴⁰ The need to solve real-world challenges posed by e-commerce and transaction volume has become an important driver of innovation for these companies. Figure 1 compares Alibaba’s transaction volume on Singles Day with that of all online sales on Cyber Monday:

36. Kapron, “Finance Disrupted: How digital finance is changing the way 1.3 billion people bank and the implications for US-China relations.”

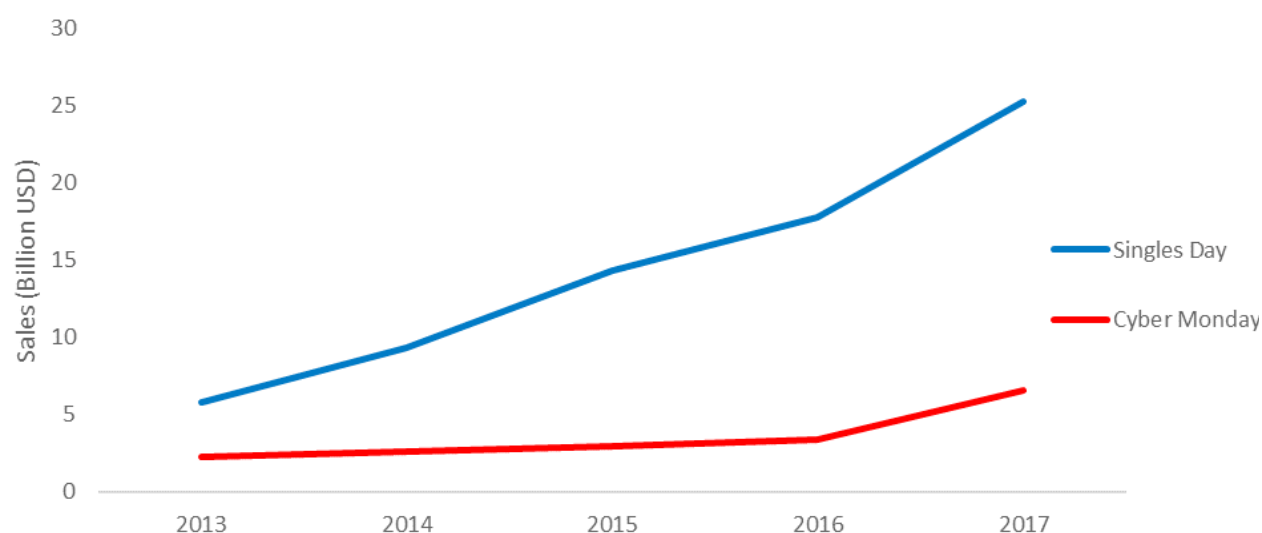
37. Interview with Paul Triolo, December 2017.

38. He Wei, “Alibaba’s Singles Day Breaks Records,” *China Daily*, November 13, 2017.

39. Tencent, Annual Report, 2016, <https://www.tencent.com/mobile/en-us/articles/17000341491836558.pdf>.

40. VISA, <https://usa.visa.com/run-your-business/small-business-tools/retail.html>.

Figure 1. Comparison of Alibaba Sales Volume on Singles Day to the Total Digital Sales on Cyber Monday, 2013–2017



Source: Statista, “Alibaba’s Gross Merchandise Volume on Singles Day from 2011 to 2017,” Adobe Digital Index Online Shopping, 2013–2017

AI is also changing Alibaba’s internet finance business. While not publicly touted much outside the company, the company is using AI across different business processes from handling massive transactions on peak shopping days to logistics and customer service.⁴¹ A coming CSIS paper will explore the use of AI in different sectors in China in more depth.

An added bonus, the bundling or integration of data from multiple sources on Chinese fintech platforms also creates advantages for fraud prevention. With visibility into the entire purchasing history and geolocation of users, Ant Financial, for example, can identify and alert users of fraud much earlier than financial counterparts in the United States. In contrast, U.S. fintech is unbundled, with users employing many different apps for functions like digital payments (Venmo or Paypal) and online investing (Robo Advisor or Charles Schwab) for discrete transactions.

But is China fintech disrupting China’s market because of deep innovation or is it simply due to a combination of external factors that do not translate globally? Three main conditions have contributed to the rise in fintech in China: first, many people have smart phones but not credit cards, and therefore made a seamless jump to mobile payments. Second, there’s an absence of regulations. And third, a large underserved population exists (consumer finance and small business) when it comes to lending.⁴² In other words, the argument could be made that the combination of technology but also user demographics and regulation have enabled the bundling model to succeed in China’s fintech sector. Peer-to-peer (P2P) lending in China may

41. Interview at Alibaba in Hangzhou, December 2017.

42. Martin Chorzempa, “P2P Series Part 1: Peering into China’s Growing Peer-to-Peer Lending Market,” Peterson Institute for International Economics, June 26, 2017,

<https://piie.com/blogs/china-economic-watch/p2p-series-part-1-peering-chinas-growing-peer-peer-lending-market>.

have flourished specifically due to the absence of regulation and consumer need in underserved areas and small business loans.⁴³

To be sure, Alibaba and Tencent are making a strong global push, focused on extending internet finance to large numbers of Chinese tourists abroad. Ant Financial has investments in local fintech companies across Southeast Asia and is accepted in over 26 countries across Europe. Ant Financial fought hard to win U.S. approval to acquire Moneygram, before it was ultimately rejected by the Committee on Foreign Investment in the United States (CFIUS). Wechat Pay launched its service in Europe in July.⁴⁴

But there are several reasons why these companies are not likely to capture U.S. or global markets in ways that come close to their success inside China. First, Facebook and others in the United States have tried bundling digital payment functions into Facebook Messenger but has not met the same scale and ubiquity as we see in China. Second, regulation has also prevented bundling in fintech in the United States. In the United States, integration of banking and commerce is largely prohibited.⁴⁵ There are also growing antitrust concerns over Amazon and “one-stop shopping.” In the EU, regulators led Google to change its shopping service in September given conflicts over having search and shopping within a single company. These factors may explain why China Alibaba and Tencent are major players in finance while internet companies in the United States and Europe are not.

Third, U.S. scrutiny over Chinese investment in the United States will also make things difficult for Ant Financial’s global ambitions in the US. Policymakers in Washington worried that Ant Financial’s bid for Moneygram would give the Chinese government access to personal information data.⁴⁶ Ant Financial states that data would be stored in the United States, but the climate is set to get more tough as draft legislation to reform CFIUS gains momentum. Political tension over Chinese deals like Ant Financial will only get worse as part of a much bigger play by Washington to bring a more confrontational policy approach to dealing with Beijing in 2018.

Fourth, Chinese online payment apps are nearly prohibitive for foreigners because they do not have a national identification card (*shenfenzheng*). There are dozens of blog entries devoted to foreigner struggles using Taobao, Alipay, and Wechat Wallet.⁴⁷ One entry begins: “How I managed to set up wechat wallet outside of China” and asks “if anyone has found other ways to transfer money to their WeChat Wallet.”⁴⁸ As the blogger’s last question suggests, there is

43. Chorzempa, <https://piie.com/blogs/china-economic-watch/p2p-series-part-1-peering-chinas-growing-peer-peer-lending-market>.

44. “Alipay vs Wechat: Challenges and Strategies of Two Payment Giants Going Global,” *Technode*, August 8, 2017, <https://technode.com/2017/08/18/alipay-vs-wechat-challenges-and-strategies-of-two-payment-giants-going-global/>.

45. U.S. Department of Justice, “Should Banking Be Kept Separate from Commerce?,” Economic Analysis Discussion Paper, <https://www.justice.gov/atr/should-banking-be-kept-separate-commerce>.

46. Shai Oster, “U.S. Scrutiny Drives Away Chinese Tech Investors,” *The Information*, December 6, 2017, <https://www.theinformation.com/u-s-scrutiny-drives-away-chinese-tech-investors>.

47. “How to set up Alipay as a foreigner,” *The Beijinger*, July 10, 2015, <https://www.thebeijinger.com/blog/2015/07/10/yes-foreigners-can-use-alipay-heres-how>.

48. “Wechat Wallet for Foreigners,” *Medium*, <https://medium.com/@hvandenbergh/wechat-wallet-for-foreigners-cb0416ee3898>.

not currently a way to top up the balance in Wechat without having funds sent from a user with a Chinese bank account in Chinese currency. Wechat will not succeed serving more than just Chinese tourists traveling abroad unless it changes these features to make it more user-friendly for global users.

However, perhaps a better way to look at the issue is how Chinese fintech companies will increasingly influence those in the West. U.S. companies are increasingly adopting lessons from Chinese internet companies to help develop their products at home. For example, one CEO commented after a trip to China, “learning from a country that drives 11x the amount of mobile payments than the U.S. demonstrates how far we still have to go in creating efficiency, speed and autonomy in the transaction and execution portion of retail.”⁴⁹ A Denver-based company with a mobile app for cash rebates called China’s internet scene “futuristic.” The second-highest-ranked shopping app in the United States, Wish, looked to Alibaba’s Taobao as a model.⁵⁰

II. Musical.ly

Tens of millions of teens in the United States are obsessed with the “selfie” video-streaming app Musical.ly, but not many realize that the app was built by a team of Chinese engineers in Shanghai. Musical.ly is China’s first digital product company to make it in the United States. People post videos between 15 seconds and two minutes—essentially the digital version of “singing into a hairbrush.” Live streaming video apps are not new in or outside China; it is a multibillion-dollar industry in China with hundreds of apps where people video themselves doing mundane activities like eating and driving. Yet, Musical.ly has created a cultural phenomenon with a package of features allowing users to manipulate and filter streams, share clips, and interact with an online community of thousands of global users. Celebrities are using it to promote songs as a new source of revenue. The Dallas Mavericks stream behind-the-scenes videos of their practices to thousands of viewers.⁵¹

To date, Musical.ly is an example of a Chinese internet company either succeeding in China or overseas, but not both.⁵² Despite its Chinese origins, the company did not do well in China, which is surprising given the popularity of live streaming in China. But this could change following its acquisition the Chinese content giant Bytedance for \$800 million. Bytedance brings to Musical.ly over 120 million daily users in China; Musical.ly brings not only an overseas following but an understanding of how to capture attention of global audiences. Therefore, the company will be important to watch at the forefront of global convergence.

49. Hans Tung, “How Amazon and Whole Foods Can Learn from China—Field Notes from China’s Retail Revolution,” <https://hans.vc/china-supermarkets/>.

50. Hans Tung, “Bytedance-Musical.ly Merger Ushers in New Age for Content Companies,” December 16, 2017, <https://hans.vc/bytedance-musical-ly-merger/>.

51. Mickey Rapkin, “The Social Media Platform That Has Gen Z Obsessed,” *Wall Street Journal*, November 1, 2017, <https://www.wsj.com/articles/the-social-media-platform-that-has-gen-z-obsessed-1509586335>.

52. Liu Yiming, *Caijing Magazine*, September 5, 2016, <http://tech.sina.com.cn/i/2016-09-05/doc-ifxvqcts9494640.shtml>.

III. Huawei

It is no surprise that Huawei was one of only four Chinese companies specifically identified by in the Cyberspace Administration of China (CAC) *Qiushi* article calling for making China's internet industry "big and strong" as well as "globally influential." (see Section II). We place it in the disruptor category as an example of innovation in its management, strategy, and business model, which has enabled to evolve into a global manufacturing giant with no real Western counterpart.

Huawei is a powerhouse, with businesses spanning wireless telecommunications infrastructure, enterprise networking, cloud, and consumer devices. Its top rival Ericsson in the 5G race does not make handsets and is struggling financially. Meanwhile, in 2016 Huawei revenue topped \$75.1 billion, up 32 percent from 2015.⁵³ In September, Huawei surpassed Apple in global smartphone sales, becoming second only to Samsung.

Yet underpinning Huawei's success is a unique approach to strategy and leadership. Some parts of the story are well documented, such as its rotating CEO model where deputy chairmen serve six-month stints.⁵⁴ Less known is how its ability to continuously transform itself as the company as it grew very quickly over more than 30 years. "Like changing a tire while driving very fast," according to Huang Can, head of Department of Innovation, Entrepreneurship and Strategy at Zhejiang University. No other company has shown an ability to do this in a sustained, meaningful way for this period of time.⁵⁵ This approach will make Huawei a leading player to watch, driving 5G development and shaping the next-generation internet.

The area globally where Huawei could face obstacles is its planned expansion to the U.S high-end smartphone market. Huawei is already among China's most successful international companies with 60 percent of its revenue from overseas and nearly 200,000 worldwide employees. But it is likely to have difficulty competing with Apple for U.S. consumers when it launches its Mate 10 smartphone with AT&T in February.⁵⁶ Huawei plans to spend tens of millions on branding in the United States and has been meticulously studying U.S. consumer habits in preparation for the launch. Just as we are watching to see if Muscal.ly can succeed in China, the question is whether Huawei can succeed in the US handset market. They have a steep hill to climb.

Narrative 2: "China-style" or Micro-Innovation

In addition to disruptive innovation, China excels at a kind of "China-style Innovation (中国式创新)," or micro-innovation, in which a technology or business model from the West is localized to the China market.⁵⁷ More simply, Chinese companies derive a new use from concepts or technology that already exists. Yet, this is much more than "Copy to China," but reflects a level

53. Huawei, Annual Report 2016.

54. David De Cremer and Tian Tao, "Huawei's Culture Is Key to Its Success," *Harvard Business Review*, May 2015, <https://hbr.org/2015/06/huaweis-culture-is-the-key-to-its-success>.

55. Interview with Dr. Huang in Zhejiang; and Wu Xiaobo, Huang Can, and Johann Peter Murmann, 华为管理变革 ("The Management Transformation of Huawei"), (Beijing: Zhongxin Publishing Company, October 2017).

56. Juro Osawa, "Inside Huawei's Plans to Challenge Apple in the U.S.," *The Information*, December 27, 2017, https://www.theinformation.com/inside-huaweis-plans-to-challenge-apple-in-the-u-s?eu=c_GYcDDybbrs_HSbdr5Rw#-continue-reading.

57. Liu Yiming. Interview at Ali Research Institute, Beijing, December 2017.

of creativity and entrepreneurship necessary to stand out in an intensely competitive market. There are dozens of examples of breakthroughs in applications, use cases, and functions, even without innovation in core technology. As the Chinese magazine *Caijing* put it, “China takes advancements in science and technology and matures them, reduces costs, and popularizes them so as to transform people’s everyday lives. This is the gene of China’s internet.”⁵⁸ We look at two cases of micro-innovation in China in this section: super apps and dockless bike sharing, followed by an assessment of their prospects for cross-border convergence.

I. “Dynamic Integration” or Super Apps

There is a concept in Chinese that can be translated as “dynamic integration” (有机结合),⁵⁹ meaning that dozens of different services or sub-apps come together under a single platform. Chinese internet companies have pioneered the development of these kinds of new platforms and ecosystems, drawing on Western company experience but tailoring applications to Chinese consumer preferences. These platforms offer many examples of business model innovation that are unique to China and the Chinese marketplace, where credit card usage and logistics infrastructures were hugely underdeveloped and large and small companies have been able to develop compelling applications. Here the online-to-offline worlds come crashing together, with payments, logistics, and customer service all playing unique roles in the China context. Understanding the confluence of these trends within China is critical to understanding how Chinese companies will be able to operate globally and the challenges they will face.

These Chinese mobile apps depend upon one technology in particular that was invented outside China: the QR code. Invented in Japan, QR codes have become ubiquitous in China as the key enabling technology for bridging the digital and real economy.⁶⁰ QR codes consist of black square grids read by an imaging device, essentially a “hyperlink and bookmark with the physical world.”⁶¹

The classic example of a Chinese super app is Tencent’s Weixin (Wechat, as it is known in English).⁶² It is China’s number-one communication tool with nearly 900 million people using it daily to send chats and share group updates along with facilitating access to myriad services, including entertainment, ride-hailing, food delivery, and more—all accessible without leaving the platform. At the heart of WeChat is the capability to transfer money and payments between users and between users and merchants (as discussed in our section above on fintech). Users in China spend on average 1.5 hours a day on the app in contrast to an average of 50 minutes on Facebook’s suite of apps combined (Facebook, Instagram, and Messenger).⁶³

58. Liu Yiming, *Caijing Magazine*.

59. Ibid.

60. Interview at Ali Research Institute, Beijing, December 2017.

61. Connie Chen, “16 Ways QR Codes Are Being Used in China,” <https://a16z.com/2017/08/11/qr-codes-in-china/>.

62. Jonah Kessel and Paul Mozur, “How China Is Changing Your Internet,” *New York Times* video, August 9, 2016, <https://www.nytimes.com/video/technology/100000004574648/china-internet-wechat.html>.

63. David Cohen, “How much time will the average person spend on social media?,” *AdWeek*, March 22, 2017, <http://www.adweek.com/digital/mediakix-time-spent-social-media-infographic/>.

Within Wechat is another layer called a Wechat Mini-program. Mini-programs allow other services to build apps within Wechat's platform. These appear and disappear without users needed to install them or take up memory. Some speculate that Wechat is on its way to becoming its own operating system. To illustrate how it works, when you go to a restaurant, you simply scan a QR code to load the program and order, without needing to install a separate app for the restaurant. The program clears from the phone after the meal.⁶⁴

Beyond WeChat, new kinds of "dynamic integration" apps are popping up every day in China that help weave the digital economy more seamlessly into daily life. The dominant one is Meituan Dianping—valued at \$30 billion this fall—which offers users food delivery, restaurant bookings and reviews, groceries, travel, plus anything from haircuts to car washes. The company began as one of the thousands of group discount clones of Groupon.

But there are also emerging companies that embody the concept internet technology to unlocking greater productivity in the real economy. A new app called Flash Delivery (闪送) is just one example. People use the app to sell and buy services for absolutely anything, using an Uber-like service for delivery. As one user described it, he left a coat in his office so used Flash Delivery to find a person to retrieve it and then deliver it to his house, all within one hour. Or when a local store sold out of a new popular seasonal ice cream, he used the app to hire a person to pick up the order (from an online store like ebay and overstock.com) then deliver it. The app is like a combination of Uber, Amazon, overstock, ebay, and more using the speed and efficiency of China's e-commerce logistics infrastructure, where goods can be delivered much more quickly than in the United States.

Looking at the global prospects for super apps, in the previous section we discussed the challenges companies like Wechat face becoming international players beyond offering services to Chinese tourists. Yet, China's super apps are a great illustration of how global convergence is taking place even without companies like Wechat or Meituan Dianping looking to take on the U.S. market. Facebook Messenger is looking to introduce bundled functions by offering Uber and payment within the app. Apple also appears to be playing catch-up⁶⁵ by introducing new OS features that have long been a staple of WeChat to appeal to Chinese users and improve retention. This includes QR code support and improved dictation for Chinese. More importantly, iMessage has evolved from a simple texting tool into a messaging platform that integrates with Apple Pay and other third-party applications that support navigation, e-commerce, and multimedia sharing. Since the launch of the iMessage app store in September 2016,⁶⁶ however, enthusiasm from developers has waned and the growth in new applications has shrunk. Yet the popularity of universal messaging⁶⁷ apps is unlikely to fade, demonstrating

64. Zara Zhang, "Philosophies Underlying the Wechat Mini-program," *zarazhang.com*, August 27, 2017, <https://zarazhang.com/2017/08/27/4-philosophies-underlying-the-wechat-mini-program-lessons-from-the-father-of-wechat/>.

65. "Apple plays catch up with new iOS 11 features for China—TechNode," *Technode*, <https://technode.com/2017/06/06/apple-plays-catch-up-with-new-ios-11-features-for-china-2/>.

66. Parmy Olson, "People Aren't Building Apps for Apple's iMessage Like They Used To," *Forbes*, March 17, 2017, <https://www.forbes.com/sites/parmyolson/2017/03/17/apple-imessage-apps-growth/#9a331362a402>.

67. David Pierce, "Screw Texting. It's Time to Pick a Universal Messaging App," *Wired*, August 16, 2015, <https://www.wired.com/2015/08/time-to-ditch-texting/>.

how WeChat's design decisions have entered the global marketplace of ideas and influenced users' expectations for integrated services through a single platform.

II. Dockless Bike Sharing

There is a lot of buzz around Chinese dockless bike-sharing companies. A handful of Chinese companies such as ofo and Mobike took the concept of bike sharing and introduced a key micro-innovation: riders can pick up and leave bikes anywhere. The system embodies the OMO concept, using mobile apps with digital payment and bikes outfitted with smart locks and GPS tracking. Dockless bikes have run into problems, such as tangled mountains of bikes in Chinese cities—but the concept is a true example of micro-innovating originating in China. Mobike and ofo currently average 30–35 million trips per day in China alone, at least five times higher than Uber's average rides per day, globally, in 2016 (5.5 million).^{68, 69}

It also illustrates global convergence for two reasons. First, these Chinese bike-share companies are now leading to imitations here in the United States. LimeBike is a California-based dockless bicycle-sharing company founded in 2017 following in the footsteps of ofo and Mobike. Second, companies like ofo were created with the goal of expanding internationally from the outset. The founders chose the name "ofo" thinking that it would not need to be translated into different languages. The logo of a rider and two wheels is similarly meant to be a universal image. Even the payment system is set up for global use. The company uses an international payments company called Ayden for users in the United States, not Alipay. This is important for making ofo a global company given difficulties for foreigners using Alipay and Wechat Pay, as discussed earlier. The bike-sharing startup environment in China has benefited from significant cross-pollination with Silicon Valley in terms of staffing. Members of the so-called "Kalanick mafia," a group of U.S.-educated Chinese nationals and former Uber employees originally involved in Uber's China operations, serve in key positions at both Mobike and ofo, where they continue to implement many of Kalanick's business philosophies and management practices in those firms.⁷⁰

Narrative 3: "Copy and Paste" or "Copy to China"

During the first phase of China's internet economy (roughly 1985–2000), Chinese companies were notorious for copying the business models of successful U.S. and European companies. Fake Chinese Apple stores and clones of Groupon, Facebook, Tumblr, and beyond reinforced China's reputation as a country of "copycat" instead of innovation. This "copy to China" or "copy and paste" storyline is best illustrated by Groupon's experience. In 2011, there were roughly 5,000 "Groupon clones" in China's group buying industry, 3,000 of which sprang up in the first half of that year alone.⁷¹

68. "China's bicycle-sharing giants are still trying to make money," *The Economist* November 25, 2017, <https://www.economist.com/news/business/21731675-one-answer-would-be-ofo-and-mobike-merge-chinas-bicycle-sharing-giants-are-still-trying>.

69. Fitz Tepper, "Uber has completed 2 billion rides," *TechCrunch*, July 18, 2016, <https://techcrunch.com/2016/07/18/uber-has-completed-2-billion-rides/>.

70. Jordon Schreiber, "Ex-Head of Mobile at Mobike Max Zhou on Dockless Bikesharing in China," China Econ Talk Podcast. December 22, 2017.

71. Steve Millward, "China's Most Audacious Groupon Clone on the Brink, 80% of Staff Fired," *Techinasia*, September 28,

It is not just foreign companies that struggle with business model imitation. Chinese entrepreneurs say that their number-one obstacle to innovation is IP theft. As *Caijing* describes it: "In Silicon Valley, when a new business model emerges, no more than three to five companies in the US will be working on it...But in China, whenever a new idea emerges, as long as the concept is feasible, there will be dozens and hundreds of imitators, making the competitive environment extremely poor."⁷²

Meanwhile, stories of a start-up bubble in China also abound with "hyper-competition backed by the huge amount of capital available to prop up new companies."⁷³ Even JD.com founder called out problems with start-ups overstating their bases and exaggerating proprietary technology.⁷⁴

Even at the platform level, China has already seen the rise and fall of large tech firms trying to emulate aspects of U.S. tech company diversification into other sectors. LeEco is a good example of this, building an entertainment juggernaut before branching into electric vehicles, which then fell spectacularly under a mountain of debt and unfulfilled promises, including major investments in the United States for plants and personnel.

The case of Didi is illustrative. Didi Chuxing came to be China's dominant ride-sharing company through imitation, politicking, and a savvy acquisition strategy that eliminated Uber, its foreign rival, from the market. Didi is now the world's second-largest online transaction platform (after Alibaba) with a valuation of \$56 billion, with nearly 400 million users in China.⁷⁵ One could make the argument that Didi is an example of micro-innovation because it managed to localize Uber to make it more appealing in China. It not only integrated regular taxi services into its app (granted, Didi was not able to do this as a foreign company), but also integrated additional features like games and social functions such as wechat and qq. As one user, who is familiar with Uber's Beijing-based operations, put it, "some people just found Didi more familiar and fun to use."⁷⁶

But the reality is that Uber simply could not sustain operations in China after government regulations made it virtually impossible for them to survive.⁷⁷ Then, the government approved the merger of its two Chinese rivals, Didi Dache and Kuaidi Dache, for \$6 billion, creating a rising Chinese tech national champion that moved to acquire Uber at the perfect moment.

2011, <https://www.techinasia.com/groupon-china-clone-tuanbao>.

72. 刘一鸣 (Liu Yiming), "中国互联网公司进入大航海时代:海外扩张成下一轮机遇" ("China's Internet Companies Enter the Era of Big Sails: The Next Round of Opportunities for Overseas Expansion") 财经, September 5, 2016, <http://tech.sina.com.cn/i/2016-09-05/doc-ifyvqcts9494640.shtml>.

73. Ben Yunmo Wang, "Frenzy and Froth in China's Start-up Scene," *Foreign Policy*, October 15, 2015, <https://foreignpolicy.com/2015/10/15/froth-and-frenzy-in-chinas-start-up-scene-is-the-bubble-popping/>.

74. 刘强东炮轰互联网泡沫 行业疯狂会出问题 ("Liu Qiangdong Shelling the Crazy Problem of the Internet Industry Bubble"), 财新网 (*Caixin Online*), September 22, 2015, <http://companies.caixin.com/2015-09-22/100854782.html>, as cited in Wang, "Frenzy and Froth in China's Start-up Scene."

75. Didi Chuxing website, accessed December 17, 2017, <http://didichuxing.com/en/aboutus/products>.

76. Interview with user, December 2017.

77. William C. Kirby, "The Real Reason Uber Is Giving Up in China," *Harvard Business Review* website, August 2, 2017.

The question looking ahead is: Can Didi capture global market share and compete with Uber when the playing field is level outside of China? Didi has made investments in emerging market companies including Grab in Southeast Asia, Ola in India, and 99 in Brazil, while stating that international expansion is a core policy. Having billions in cash and success inside China alone do not themselves translate to global competitiveness. Didi will have to focus on localizing to overseas markets and address a host of human-resource and technology hurdles.⁷⁸

78. 滴滴真的要去海外做打车生意了, 它还有多少机会? ("How large realistically are opportunities for Didi to expand its business overseas?"), 腾讯科技 (Tencent Technology), February 22, 2017, <http://tech.qq.com/a/20170222/013513.htm>; and Yunan Zhang, "China's Didi in Talks to Buy Brazil's 99, Threatening Uber," The Information, December 21 2017, <https://www.theinformation.com/chinas-didi-in-talks-to-buy-brazils-99-threatening-uber>.

Policy Challenges for Global Convergence

The administration of President Xi has stated in numerous forums that government policy should balance economic development with information security.⁷⁹ The emphasis on boosting the digital economy as a source of growth comes alongside the construction of legal framework for strengthening cybersecurity, with the cybersecurity law (CSL) as the centerpiece.⁸⁰ The CSL applies to both content and critical information infrastructure. For Chinese internet companies, the CSL and its associated measures can be seen as an enabler and also an obstacle for China's digital economy. We look at both, with a focus on two obstacles: restrictions on data flows and increasing localization of ICT in China.

CSL AS AN ENABLER OF CHINA'S DIGITAL ECONOMY

The implication of President Xi's slogan "Without cybersecurity, there is no national security; without informatization, there is no modernization"⁸¹ is that the digital economy cannot advance without security. Indeed, one senior executive at a leading Chinese ICT company said that the CSL was meant to "get people to trust the internet."⁸²

79. 《网络安全法》施行前夕，国家网信办网络安全协调局负责人答记者问 (On the Eve of the Cybersecurity Law Taking Effect, National Cyberspace Administration of China Cybersecurity Division Director Answers Questions), May 31, 2015, <http://mp.weixin.qq.com/s/n6OPhHkN6DAMYjBGLAlVcA>.

80. Paul Triolo, Samm Sacks, Graham Webster, and Rogier Creemers, "China's Cybersecurity Law One Year On," November 30, 2017, <https://www.newamerica.org/cybersecurity-initiative/digichina/blog/chinas-cybersecurity-law-one-year/>; and Samm Sacks, "China's Emerging Cyber Governance Framework," <https://www.csis.org/chinas-emerging-cyber-governance-system>.

81. 中央网络安全和信息化领导小组第一次会议召开 习近平发表重要讲话 (Xi Jinping's Speech Delivered at the First Meeting of the Central Leading Small Group for Cybersecurity and Informatization), February 27, 2007, http://www.cac.gov.cn/2014-02/27/c_133148354.htm

82. Interview with leading Chinese ICT company executive in Beijing, December 2017.

Mobile fraud and cybercrime is a genuine concern in China. A case of an incoming college student persuaded to transfer tuition funds as part of an online scam caused an uproar in China last year. User concerns over cyber fraud represented a major obstacle for e-commerce when Alibaba first launched Taobao. In fact, Alibaba created Alipay in 2004 to order to address user concerns with trusting online transactions thousands of miles away.⁸³ Alibaba designed Alipay to be tailored to customer preferences around e-commerce so sellers would not be paid until buyers were satisfied with the purchase.⁸⁴ Chinese government officials have themselves stated that the development of industry standards and best practices for cybersecurity lags behind advancements in technology.⁸⁵ This is a main driver of the rapid buildout of a new legal and policy regime around cybersecurity. At the same, China's use of outdated (and often pirated) software without advanced security features remains rampant, raising questions about the call for laws as the solution for cybercrime.

CHINA'S EMERGING DATA REGIME AND DATA TRANSFER

At the moment when Chinese and Western internet companies are embarking on global convergence in their digital economies, new rules around how data is transferred across borders threatens to fragment market opportunities and R&D.

According to article 37 of the CSL: "Personal information and other important data gathered or produced by critical information infrastructure (CII) operators during operations within the mainland territory of the People's Republic of China, shall store it within mainland China."

In practical terms, the provision means that data falling in scope would need to be stored inside China and at certain levels of security, and need to undergo procedures before being transferred out of China.⁸⁶ The definition of many key terms of the provision are still under review, but as things now stand, the rule would apply to anyone who owns and manages an IT network, including e-commerce.⁸⁷

Influential stakeholders in the Chinese system are advocating for alignment international practices and global interoperability when it comes to data flows. From national tech champions like Alibaba seeking global markets, to Chinese financial institutions facilitating global transactions, cross-border data flows are a core operational reality. For example, Song Fei, vice president of the Ali Research Institute, gave a speech in 2015 (when data localization became more prominent in pending legislation) stating the importance of cross-border data flows for global commerce, stating that Alibaba's exports generated over 21 million orders from 200 countries and regions; 30 million Chinese users imported goods.⁸⁸ Dr. Hong

83. Interview at Ali Research in Beijing, December 2017.

84. KapronAsia, https://www.uscc.gov/sites/default/files/Kapron_USCC%20Testimony_22June2017.pdf.

85. Interviews in Beijing, December 2017.

86. Triolo et al., "China's Cybersecurity Law One Year On,"

87. Samm Sacks, "China's Cybersecurity Law Takes Effect: What to Expect," Lawfare Blog, June 1, 2017, <https://www.lawfareblog.com/chinas-cybersecurity-law-takes-effect-what-expect>.

88. 国内首次“跨境数据流动”研讨会在京举 (China's First Cross-Border Data Flow Seminar Held in Beijing), Ali Research, <http://www.aliresearch.com/blog/article/detail/id/20701.html>, November 19, 2015.

Yanqing is an example of an authoritative figure helping to shape policy as the lead for the personal data protection project for TC260 and is deputy head of the task force on the cross-border data transfer guidelines. He writes, "A fundamental consensus has emerged today that data naturally flows across national borders, that data flows produce value, and that data flows can lead to flows of technology, capital, and talent. Therefore, data flows are the norm, and circumstances where flows are limited are the exception."⁸⁹ Yet, as government officials continue to debate how these measures of CSL will be implemented, the final outcome of these discussions is not yet clear.

CHINA'S EMERGING DATA REGIME AND DATA PRIVACY

As China's data regime takes shape, it is not just policies around data transfer that will impact China's digital economy, but also how the leadership approaches data privacy. This will have important implications for BATJ's global expansion, in terms of users' concerns about how their personal information will be handled, as well as meeting data privacy and protection requirements in markets like the European Union.

To be sure, Chinese users have fewer concerns about data privacy than in the West, and are often more willing to share data for better access to services and applications.⁹⁰ Weaker privacy protections in China may be a factor in the growth of Chinese internet firms, which rely on advertising similar to the U.S. model. Yet, there is a tug-of-war inside China over data privacy. Debates are thriving between proponents of stricter security controls for personal information and those more focused on accelerating the development of AI and machine-learning fields. In the most recent example, user outcry led Ant Financial to issue an apology after an Alipay default option allowed its credit scoring system to access user data.⁹¹ Cai Xiongshan, chief researcher of Tencent Research Center's Institute of Law, articulates the challenge in "Preliminary Exploration of Artificial Intelligence and Legal Issues." He writes that there are unresolved questions regarding whether businesses or users have ownership of data. He sees a conflict between protecting national security and personal data that needs deeper research.⁹² Ali Research Institute has also been involved in providing input to the government on securing personal information data.⁹³ Experts from Alibaba and others in industry are involved in ongoing discussion with regulators on shaping standards for the kind of data that can be collected from customers, storage, handling, and product user consent agreements. It is not surprising that Alibaba and Tencent are driving a lot of debate inside China on data

89. Hong Yanqing, "The Cross Border Data Flows Security Assessment: An important part of protecting China's basic strategic resource," Yale Law School Paul Tsai China Center, June 20, 2017, https://law.yale.edu/system/files/area/center/china/document/dataflowssecurity_final.pdf.

90. Email with Liza Lin, *Wall Street Journal* correspondent from Shanghai, December 21, 2017.

91. Xinhua News, "China Focus: Ant Financial apologizes for online credit trickery," January 4, 2018.

92. 蔡雄山 (Cai Xiongshan), 人工智能与法律问题初探 ("Artificial Intelligence Legal Challenges Preliminary Assessment"), unpublished draft cited with permission of the author.

93. Interview at Ali Research Institute in Beijing, December 2017.

security and privacy policy given the importance of global markets to their business strategies. China's data security and privacy regime is likely to evolve over the next one to two years.

The resulting framework will be key to watch as another factor impacting the global role of China's internet companies.

LOCALIZATION PUSH IN ICT SECTORS

The CSL and its associated framework is part of a broader policy trend toward localization in China's ICT sector. There are two implications. First, a closed ecosystem in which foreign ICT companies struggle without local partners will not be conducive to true innovation and global competitiveness. Second, it also could come at a cost to China's internet companies seeking greater access to global markets, but faced with new restrictions as policymakers push for more reciprocity in treatment of foreign and Chinese companies.

To illustrate, AWS's decision in November to sell infrastructure assets to its Chinese partner is just one recent example of how foreign companies are reassessing their operations in China amid tightening regulation on data and cloud services. Government support in the form of decreased overhead costs in commodities and prices and requirements for foreign companies to have partnerships with Chinese domestic companies help China's state-owned and private sector internet companies maintain dominance. It is no surprise that local cloud firms have 80 percent of cloud revenue, for example. Western ICT companies are making more compromises in order to maintain market access.

Yet, this does not help companies like Alibaba—a growing cloud services player—become a viable international player capable of competing against AWS and Microsoft. It also opens the door for moves to restrict global expansion of Alibaba's data centers given increased tensions around digital trade and data flows. Meanwhile, equipment makers like Huawei could face more backlash in the United States at the very moment it makes a play for the global high-end smartphone market.

Conclusion

Digital economy is not just a party slogan in China. Chinese internet firms are transforming China's own economy and influencing Western economies. Looking forward this year and beyond, a number of questions remain, not just about China's own progress, but what global convergence will mean for emerging technologies and digital trade: Will Chinese firms narrow the gap needed to compete globally with players like AWS and Microsoft as they build out data centers and cloud-service offerings outside of China? Or will global ambitions for cloud stay focused on economies that look more similar to China, in places like Southeast Asia?

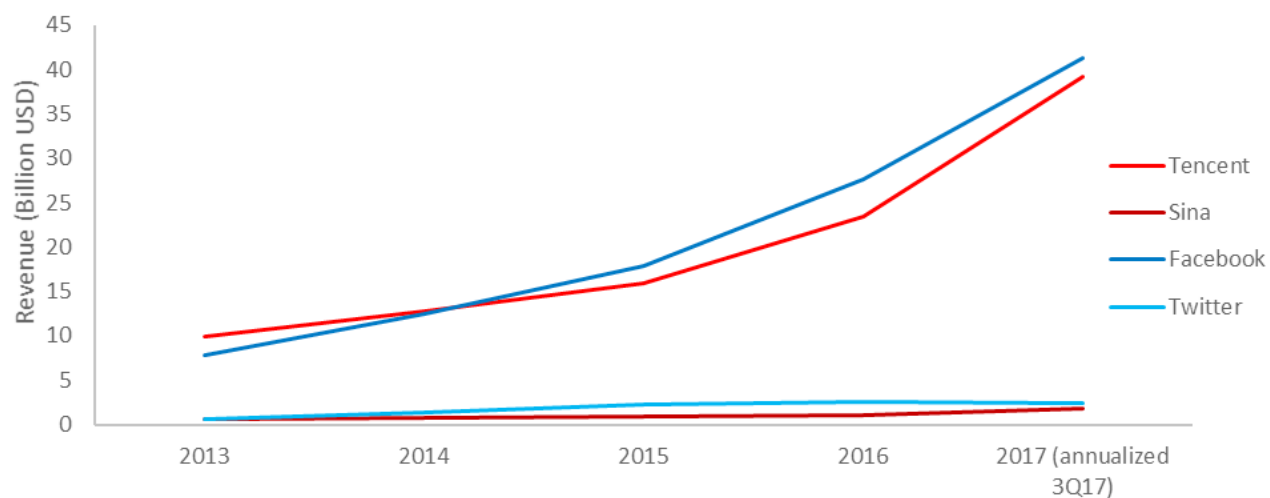
We also still do not know a lot about new major R&D efforts underway by the top Chinese companies. For example, it is still too early to tell what exactly is underway at Alibaba's new Global Research Institute—also called DAMO Academy (Discovery, Adventure, Momentum, and Outlook)—launched in October other than the tremendous amount of capital backing up these endeavors. We do know that it will focus on fields such as machine learning, natural language processing, sensing technology, and embedded systems. And consistent with the theme of global convergence, DAMO is collaborating with experts from leading data science programs at top Western universities.⁹⁴ What will come from these investments and collaborations? What is clear is that developments at companies like Alibaba, as well as Huawei and the 5G race, are going to have a growing role defining the future of the internet.

Lastly, 2018 is set to be a tense year in the bilateral U.S.-China relationship as the Trump administration moves forward with a more confrontational China policy. Coming measures from Washington will have implications for digital trade, national security, and the future of innovation convergence.

94. 阿里巴巴达摩院 (Alibaba Damo Academy), accessed on December 24, 2017, <https://baike.baidu.com/item/%E9%98%BF%E9%87%8C%E5%B7%B4%E5%B7%B4%E8%BE%BE%E6%91%A9%E9%99%A2/22159003>.

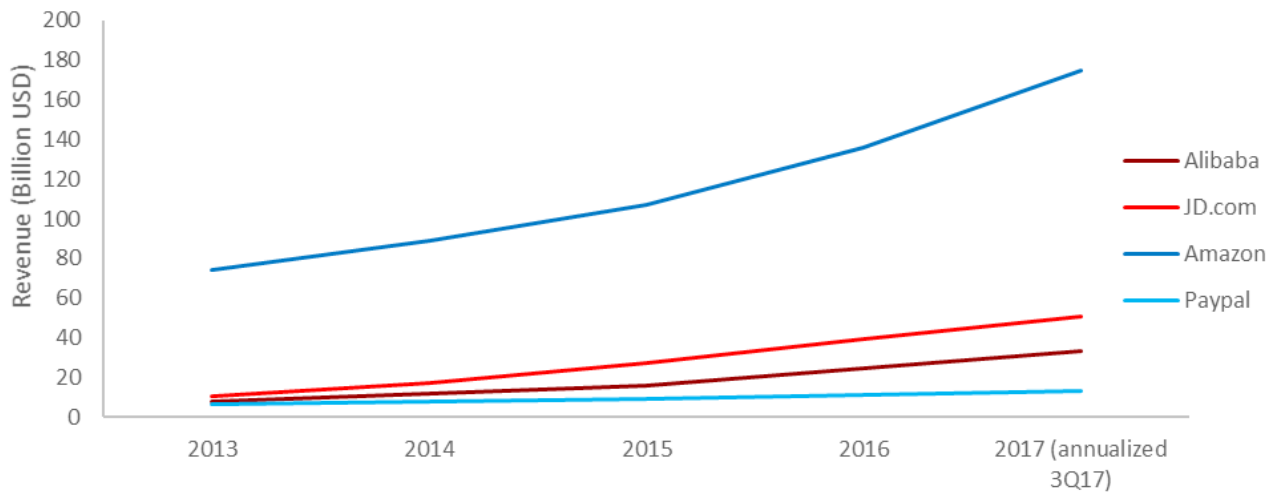
Appendix A

Figure 2. Revenue of Major U.S. and Chinese Social Media Companies, 2013–2017



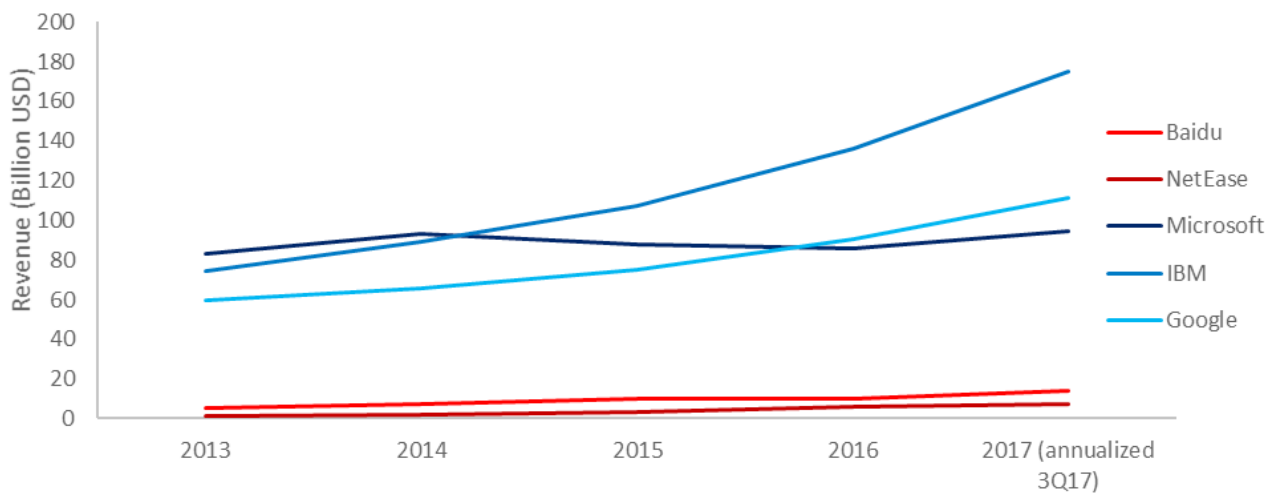
Source: Company annual and quarterly financial reports

Figure 3. Revenue of Major U.S. and Chinese E-Commerce Companies, 2013–2017



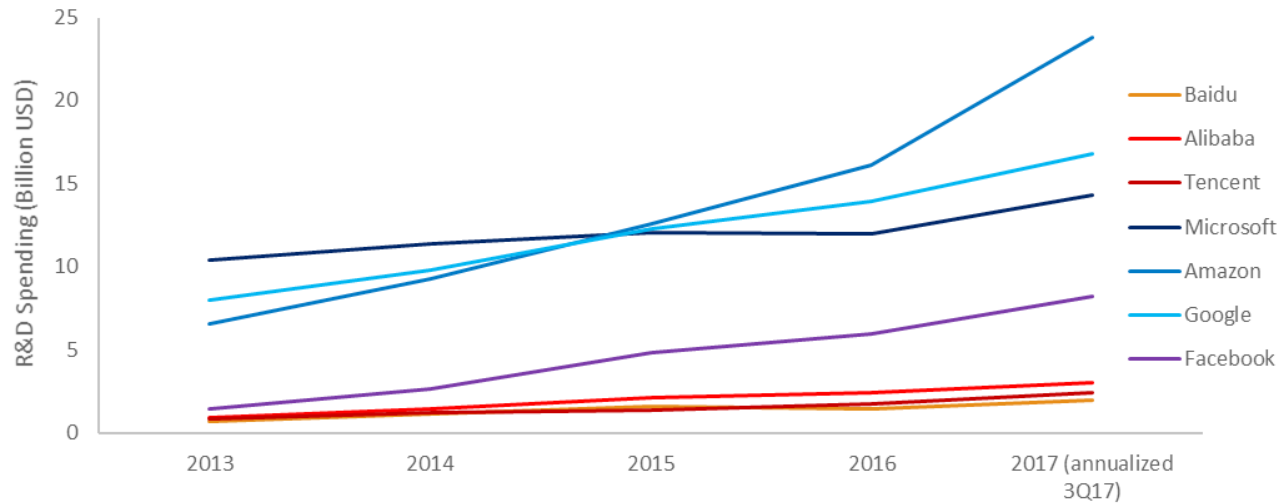
Source: Company annual and quarterly financial reports

Figure 4. Revenue of Major U.S. and Chinese Internet Service Companies, 2013–2017



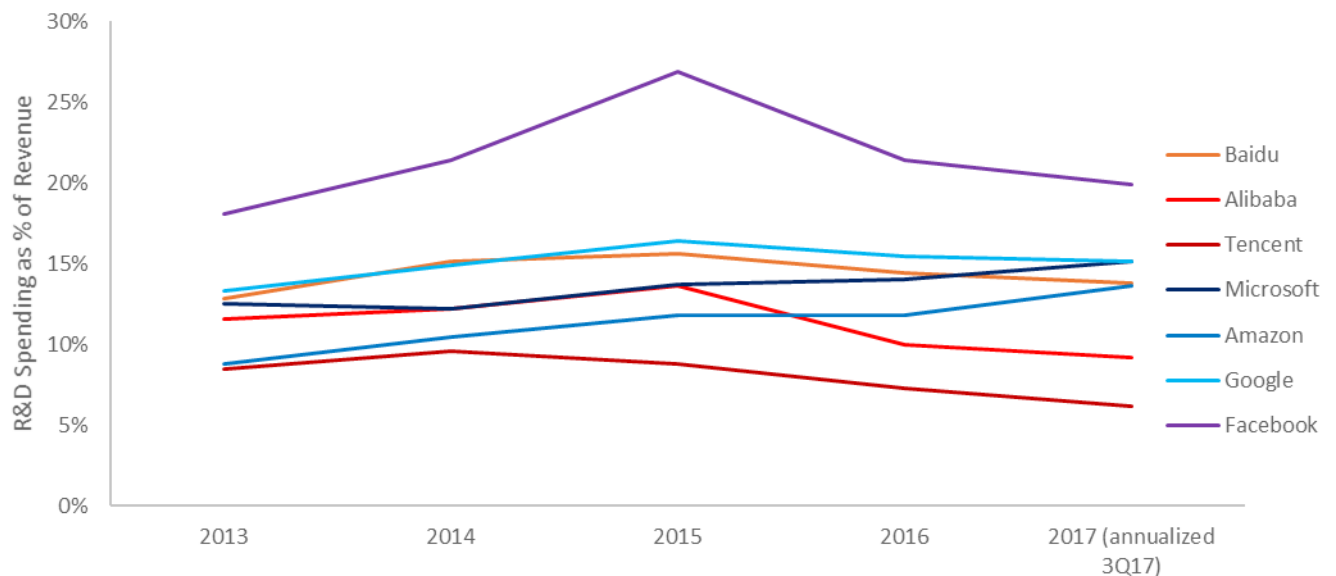
Source: Company annual and quarterly financial reports

Figure 5. R&D Spending of Major U.S. and Chinese Internet Companies, 2013–2017



Source: Company annual and quarterly financial reports

Figure 6. R&D Spending of Major U.S. and Chinese Internet Companies as a Percent of Revenue, 2013–2017



Source: Company annual and quarterly financial reports

About the Author

Samm Sacks is a senior fellow in the Technology Policy Program at CSIS. Her research focuses on innovation, cybersecurity, and emerging information and communication technology (ICT) policies globally, with a focus on China. She leads CSIS's "China Cyber Outlook," which analyzes China's evolving ICT governance system, including data flow and privacy issues, technology sector political leadership, the build-out of Chinese standards, and the global expansion of Chinese ICT companies. Before joining CSIS, she launched the industrial cyber business for Siemens in Asia, focusing on energy-sector cybersecurity. Previously, she led China technology-sector analysis at the political risk consultancy Eurasia Group. Prior to this, she worked at Booz Allen Hamilton and Defense Group Inc., where she advised senior U.S. government officials on China's science and technology (S&T) development. A former Fulbright scholar in Beijing, Ms. Sacks holds an M.A. from Yale University in international relations and a B.A. from Brown University in Chinese literature. She reads and speaks Mandarin and is a frequent contributor to print and TV media, including *Lawfare*, the *New York Times*, *Washington Post*, *Los Angeles Times*, Reuters, Fortune, Bloomberg, Nightly Business Report, C-SPAN, and BBC.

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